

Given measures of D_2 , D_3 , and C_1 , strike them out from the encoding equation.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ g_{21} & g_{22} & g_{23} & g_{24} \\ g_{31} & g_{32} & g_{33} & g_{34} \end{bmatrix} \times_{GF(2^m)} \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix} = \begin{bmatrix} D_1 \\ D_3 \\ C_2 \\ C_3 \end{bmatrix}$$

Invert to get decoding equations and recover erased data chunks D_2 and D_4 .

$$\left(\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ g_{21} & g_{22} & g_{23} & g_{24} \\ g_{31} & g_{32} & g_{33} & g_{34} \end{bmatrix} \right)^{-1} \times_{GF(2^m)} \begin{bmatrix} D_1 \\ D_3 \\ C_2 \\ C_3 \end{bmatrix} = \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix}$$

Re-encode erased parity chunk C_1 with all data chunks.

$$\begin{bmatrix} g_{11} & g_{12} & g_{13} & g_{14} \end{bmatrix} \times_{GF(2^m)} \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix} = C_1$$